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MSc in Applied Computational Science and Engineering

Independent Research Project

Final Report

Origins of Mars's Moons: Physics-Structured ML Using Asteroid Tidal-Disruption SPH Simulations

by

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Abstract

This paper investigates asteroid tidal disruption during close encounters with Mars, in which differential gravity can fragment an incoming asteroid and leave part of the debris gravitationally bound. High-fidelity smoothed-particle hydrodynamics (SPH) resolves this process but is too expensive for exhaustive parameter-space exploration. SPH outputs are therefore post-processed into fragmentation diagnostics and bound mass fraction (BMF), and a machine-learning surrogate is evaluated using leakage-safe grouped cross-validation. The deployed raw-input Random Forest achieves out-of-fold $R^2 = 0.8971$ and MAE = 0.0184 BMF. In a representative controlled slice, BMF falls from approximately 0.27 to zero as periapsis increases from 1.2 to 2.8 Mars radii, and from approximately 0.17 to zero as encounter speed increases from 0 to about 1 km/s. Periapsis is the strongest archive-wide organiser, but matched cases show that local sensitivity changes with regime: spin-induced BMF spread is generally about 0.06-0.10 below 1.5 Mars radii and reaches about 0.14-0.24 in several higher-periapsis velocity families. A sparsely sampled $10^{19.5}$ kg case gives held-out errors near 0.08 BMF despite lying within the numerical input ranges. The surrogate is therefore useful for rapid, locally supported screening and for identifying interactions worth further SPH investigation, yet it is not a universal replacement for SPH or evidence of causation by itself.

1. Introduction

1.1 Research Problem and Questions

The origin of Mars's moons Phobos and Deimos remains unresolved. One proposed pathway is that a close asteroid encounter with Mars can produce tidal disruption: differential Martian gravity stretches and fragments the incoming body, after which some debris escapes and some remains gravitationally bound. That retained material is relevant to moon-formation hypotheses because it provides a possible source population from which later dynamical evolution and accretion could act (Kegerreis et al., 2024).

High-fidelity smoothed-particle hydrodynamics (SPH) is well suited to this problem because it follows moving material parcels through large deformation, fragmentation and gravitational interaction. However, each new setup is computationally expensive, making exhaustive exploration of asteroid mass, closest approach, encounter speed, spin and numerical choices impractical. The objective of this paper is therefore not to replace SPH, but to turn an existing simulation archive into a fast decision layer that can screen candidate encounters, identify promising retained-mass cases, and indicate when a full SPH calculation is still required.

The report addresses three research questions: (1) how do orbital and physical parameters influence disruption and bound retention; (2) which inputs most strongly control fragment formation and retained debris; and (3) can machine-learning models predict outcomes across the sampled domain with explicit trust checks? The final output is a physics-structured surrogate and decision workflow rather than an unqualified black-box predictor.

1.2 Significance and Originality

The work is significant both scientifically and computationally. Scientifically, the archive probes a mechanism that could supply debris to the Martian system. Computationally, it addresses a recurring problem in simulation-led science: a detailed solver can answer one configuration well, but the cost of many configurations limits parameter-space exploration. A surrogate can map broad behaviour almost instantly once trained, allowing expensive simulations to be concentrated where they add the most value.

The main originality is the combination of physical interpretation with explicit evidence about when prediction should be trusted. The workflow converts particle-level SPH outputs into run-level targets, evaluates the surrogate on entirely held-out physical simulations, examines controlled and matched parameter families, measures local support and held-out error, and converts these diagnostics into an SPH recommendation. Machine learning is therefore used to expose archive-wide trends and local interactions without being presented as 'better physics' than SPH.

1.3 Literature Review and Modelling Rationale

Multiple scenarios have been proposed for Phobos and Deimos. Giant-impact calculations can generate debris disks around Mars (Citron et al., 2015; Canup & Salmon, 2018), transient inner moons may affect outer-disk accretion (Rosenblatt et al., 2016), and dynamical arguments have motivated a disrupted common progenitor (Bagheri et al., 2021). No single scenario yet connects debris production, capture, long-term orbital evolution and the final moons in a complete way.

Kegerreis et al. (2024) directly motivates this paper by demonstrating disruptive partial capture during close asteroid encounters with Mars. The simulations show that appreciable bound debris can arise across variations in periapsis, velocity, mass and spin, but a new configuration still requires a new simulation. This paper asks whether the archive can be learned as a structured dataset so that future configurations can be screened before SPH is launched.

SPH is a mesh-free Lagrangian method in which computational particles move with the represented material (Price, 2012). It is appropriate for violent deformation, but the derived fragment catalogue depends on post-processing. In this paper, friends-of-friends (FoF) groups nearby particles using a linking length; therefore fragment counts and fragment masses partly reflect a methodological choice as well as encounter physics (More et al., 2011). Bound mass is physically distinct: fragments are classified using their specific orbital energy relative to Mars.

The processed learning problem is small, tabular and nonlinear rather than a field-level surrogate. Tree ensembles are consequently a sensible baseline: Random Forest averages many nonlinear trees, while Gradient Boosting corrects residual errors sequentially (Breiman, 2001; Friedman, 2001). This choice is also consistent with evidence that tree-based models remain strong on typical medium-sized tabular datasets (Grinsztajn et al., 2022). Physics-informed neural networks, by contrast, are most natural when governing-equation residuals can be enforced across continuous spatial or temporal domains (Raissi et al., 2019), which is not the form of the present run-level dataset.

Alternative model structures were evaluated to test whether the zero-heavy, nonlinear BMF response could be predicted more accurately. These experiments support the interpretation of the target but are secondary to the physical results; their architectures, feature sets and grouped-validation scores are reported in Appendix B.

2. Methodology

2.1 Data Archive, Inputs, and Post-Processing

The archive contains 489 SPH simulation outputs, totalling approximately 218 GB of HDF5 particle data. Each output records particle positions, velocities, mass, density, pressure and material type. The setup parameters describe the incoming asteroid and numerical configuration; the post-processing pipeline maps those particle snapshots to fragment- and bound-retention summaries suitable for exploratory analysis and machine learning. The main physical and numerical parameters used in the workflow are summarised in Table 1.

Table 1. Main SPH setup and post-processing parameters.
Coverage is strongly uneven across the archive.

Parameter	Coverage / range	Role in the workflow
Asteroid mass	A_{1800}-A_{2100}; A_{2000} dominates	Physical mass scale; its effect on retained fraction depends on the encounter regime.
Periapsis	r_{11}-r_{30}; r_{12} and r_{16} common	Closest approach to Mars; smaller values give stronger tidal forcing.
Encounter speed	v_{00}-v_{30}; v_{00} dominates	Higher speed makes capture less favourable.
Spin	none, x, y, z, mz	Rotation rate and orientation modify disruption geometry; coverage is uneven.
Resolution	n_{50}-n_{70}; n_{65} dominates	Numerical resolution; non-n_{65} cases are sparse.
FoF linking length	0.0001-0.0126; 0.004 common	Post-processing grouping threshold; affects measured fragments but is not an encounter-physics driver.

FoF first partitions the particle debris into connected groups. For each group g, its mass is the sum of its particle masses, $M_g = \Sigma_i m_i$. The centre-of-mass position and velocity are then used to calculate the group's specific orbital energy relative to Mars. A group is classified as bound when $\varepsilon_g < 0$ and unbound when $\varepsilon_g \geq 0$. BMF is the sum of bound fragment mass divided by the original parent-asteroid mass. The supporting targets include bound fragment count, largest/average bound fragment mass, total fragment count and largest-remnant quantities. BMF is a retention proxy: it indicates material available around Mars, not proof that a stable disk or moon ultimately forms.

2.2 Learning Task and Deployed Surrogate

The primary learning task is regression of continuous BMF, preserving distinctions between small, moderate and large retained fractions. The deployed surrogate is a raw-input Random Forest using quantities available for a prospective setup together with the defined post-processing configuration. A 10% BMF threshold is applied only as a transparent prioritisation screen; it is not a separate physical criterion for moon formation.

The surrogate is used in two ways: to predict BMF for locally supported cases and to generate controlled slices in which one parameter changes while the other inputs remain fixed. Detailed comparisons with boosting, hurdle, mixture-of-experts and Gaussian-process alternatives are provided in Appendix B, Table B2.

2.3 Leakage-Safe Grouped Validation and Metrics

The archive contains repeated post-processing rows from the same physical SPH simulation, especially when different FoF linking lengths are applied. A random row-wise split would therefore leak near-duplicate physical cases between training and testing. Five-fold GroupKFold cross-validation was used instead: every row derived from one physical simulation file remains in the same fold. Each out-of-fold (OOF) prediction is consequently made by a model that did not train on that physical simulation.

Regression is assessed with R^2, mean absolute error (MAE) and root mean squared error (RMSE). R^2 measures the fraction of variation in held-out BMF explained by the model; it is not a percentage accuracy. MAE measures the average absolute error directly in BMF units, so MAE = 0.012 corresponds to about 1.2 percentage points of BMF. RMSE gives greater weight to occasional large mistakes. Classification uses balanced accuracy, F1 and ROC AUC. Global scores are necessary but not sufficient: later diagnostics test whether accuracy is locally supported and physically sensible.

2.4 Physical Interpretation Strategy

Physical interpretation combines three levels of evidence. Controlled slices vary one input while holding the others fixed to show the direction and nonlinearity of the learned response. Matched

SPH subsets hold mass, resolution, timestep and FoF configuration fixed while comparing spin or velocity families. Finally, grouped parameter-family ablation removes an original input together with related engineered proxies to estimate which information the surrogate requires on average across the archive.

These analyses answer different questions. Controlled and matched comparisons describe behaviour within a defined physical regime, whereas family ablation gives an archive-wide predictive summary. The latter does not provide effect direction, establish causation or imply that the same ranking applies to every encounter.

The setup-derived feature definitions are listed in Appendix B, Table B1, while their grouped-validation contribution is shown in Figure B1.

2.5 Trust and Decision Logic

A surrogate prediction is accepted only conditionally. The workflow checks five complementary properties: accuracy on unseen physical simulations; physically sensible response when one parameter is varied; support from nearby archive cases; stability across model families; and whether the case lies within the sampled joint domain. These checks support interpolation but explicitly lower trust for sparse, edge or extrapolative configurations.

The final decision also separates disruption from capture. Disruption severity is described by the largest-remnant fraction $f_{LR} = M_{largest}/M_{parent}$: $f_{LR} > 0.90$ is mostly intact, 0.50-0.90 indicates partial breakup, and $f_{LR} \leq 0.50$ indicates severe disruption. Retention is then assessed independently through BMF and the absolute Mars-bound mass $M_{bound} = BMF \times M_{parent}$. This two-stage screen reflects the fact that severe breakup can still leave almost no mass bound to Mars (Richardson et al., 1998; Walsh & Richardson, 2006).

3. Results

3.1 Exploratory Physical Trends

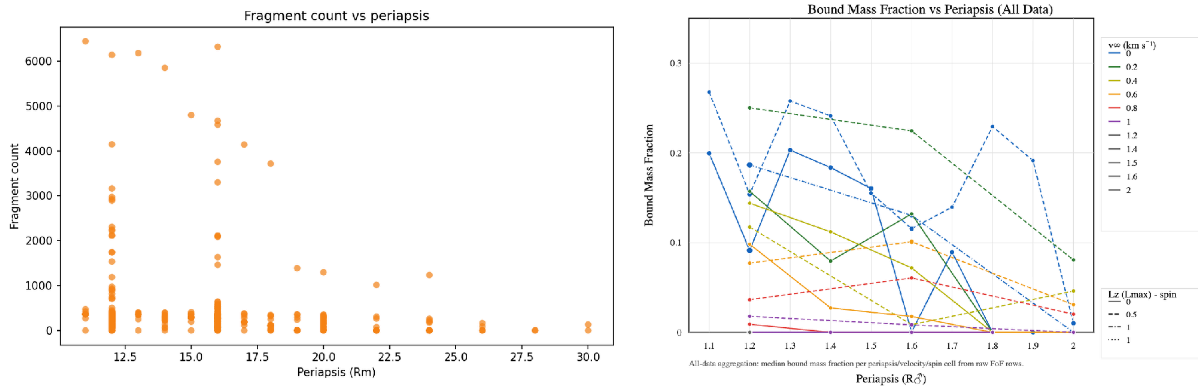

Figure 1. Exploratory SPH outcomes. (a) Extreme fragment counts occur only for close encounters, although close periapsis does not guarantee fragmentation. (b) Bound retention is generally favoured by closer and slower cases; connected lines are visual guides through sparse discrete combinations, not physical trajectories.

3.2 Eccentricity and the Retention Regime

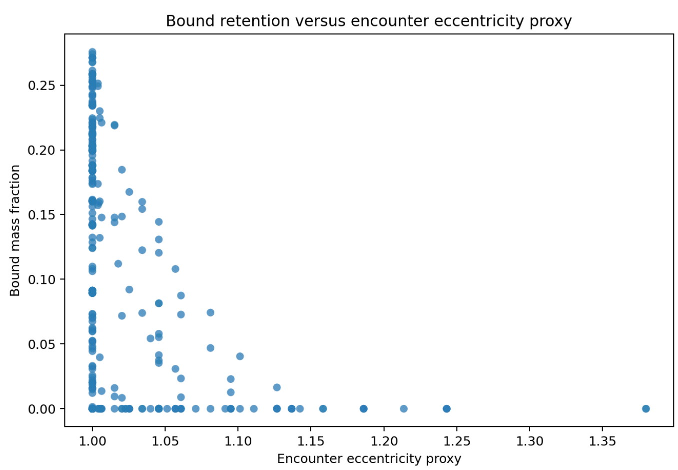

Figure 2. Bound mass fraction versus encounter eccentricity proxy. Strong and non-zero retention disappear rapidly as the encounter becomes more hyperbolic.

3.3 Surrogate Performance and Screening Value

3.4 Global and Regime-Dependent Physical Controls

3.5 Controlled Behaviour and Matched-Family Checks

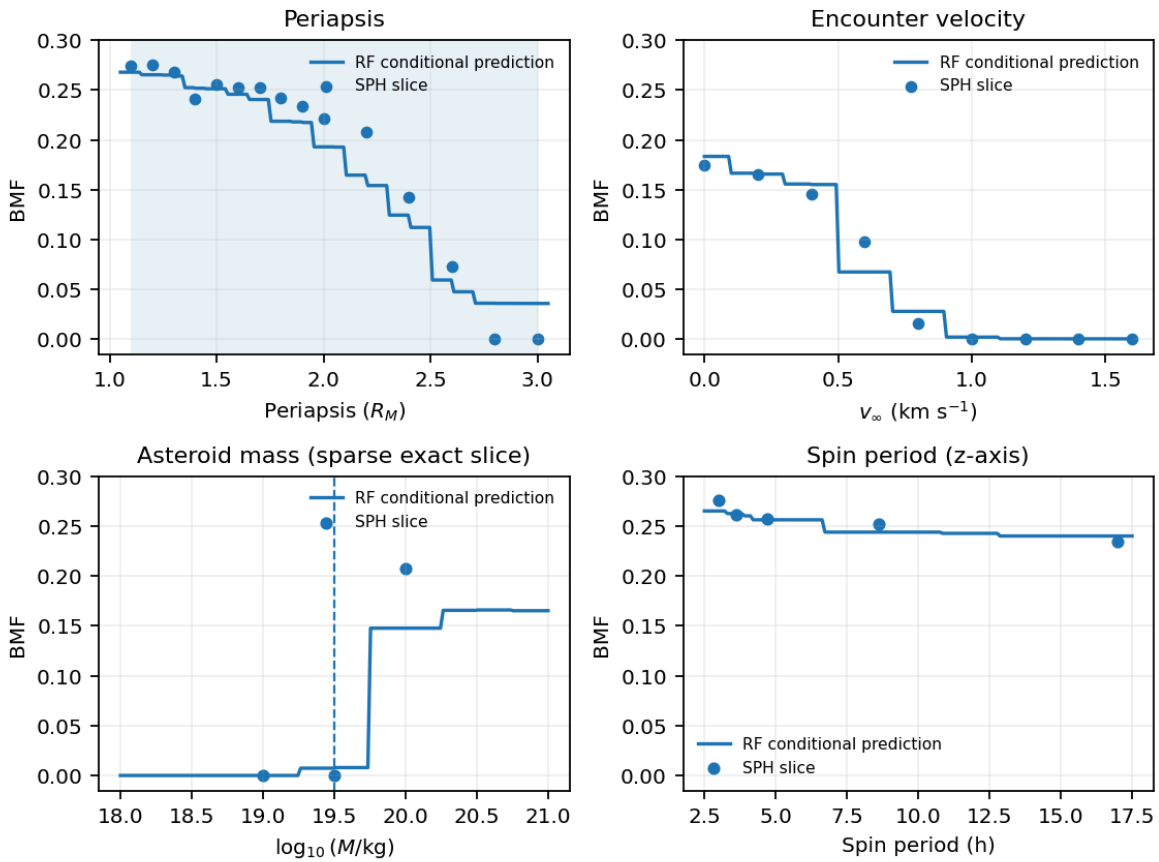

Figure 3. Representative one-parameter-at-a-time BMF diagnostics. Dots are SPH slice cases and curves are Random Forest conditional predictions. Sparse support makes the detailed mass and spin trends less secure than the periapsis and velocity responses.

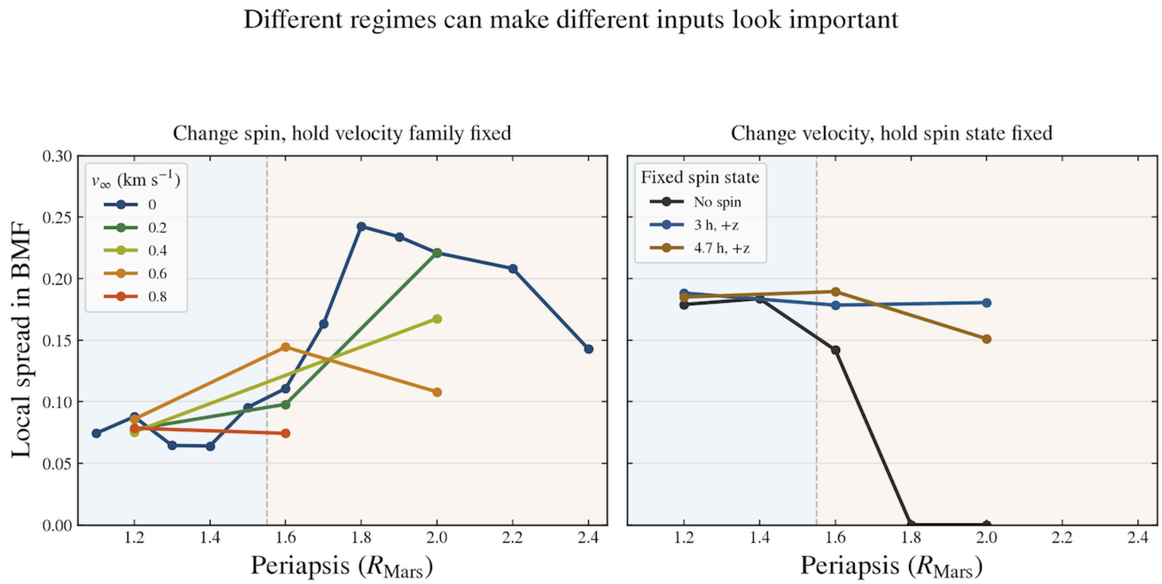

Figure 4. Regime-dependent local BMF spread in the matched A2000, n65 subset at timestep 90000 and FoF linking length 0.004. Left: spin-induced spread with velocity family fixed. Right: velocity-induced spread with spin state fixed. The spread measures local separation within this subset; it does not give effect direction or establish causation.

3.6 Coverage, Interpolation and Failure Modes

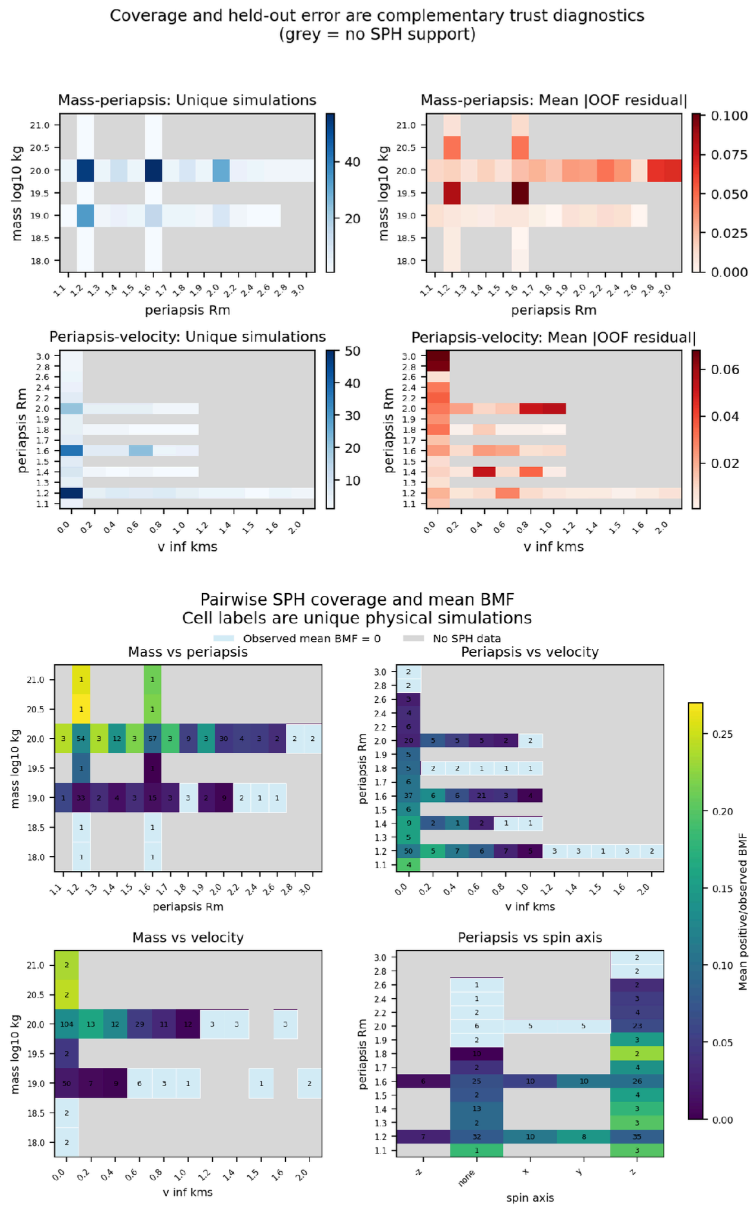

Figure 5. Trust diagnostics. Top: unique-SPH coverage and mean absolute out-of-fold error are complementary. Bottom: pairwise coverage shows that one-dimensional in-range checks can miss unsampled parameter combinations.

Table 2 quantifies this failure case for the sparsely sampled 1019.5 kg asteroid mass.

Table 2. Failure case at asteroid mass 10^19.5 kg: numerically in range, but supported by only two unique physical SPH simulations.

Periapsis	**Actual BMF**	**OOF prediction**	**Abs. error**	**Failure**
1.2	0.0913	0.0134	0.0779	Underpredicts
1.6	0.000204	0.0827	0.0825	Overpredicts

3.7 Decision-Support Interpretation

4. Discussion

5. Conclusions, Limitations and Future Work

5.1 Conclusions and Answers to the Research Questions

This research asked how orbital and physical parameters influence disruption and bound retention, which inputs most strongly control fragment formation and retained debris, and whether machine learning can predict outcomes across the sampled domain with explicit trust checks. The answers to these research questions are as follows.

Q1. How do orbital and physical parameters influence disruption and bound mass? The answer is that

Q2. Which parameters most strongly control fragment formation and retained debris? The answer is that

Q3. Can machine learning predict outcomes across the sampled domain with useful trust checks? The answer is yes, but only for locally supported combinations.

5.2 Limitations

5.3 Future Work

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Appendix A. Decision-Support Prototype

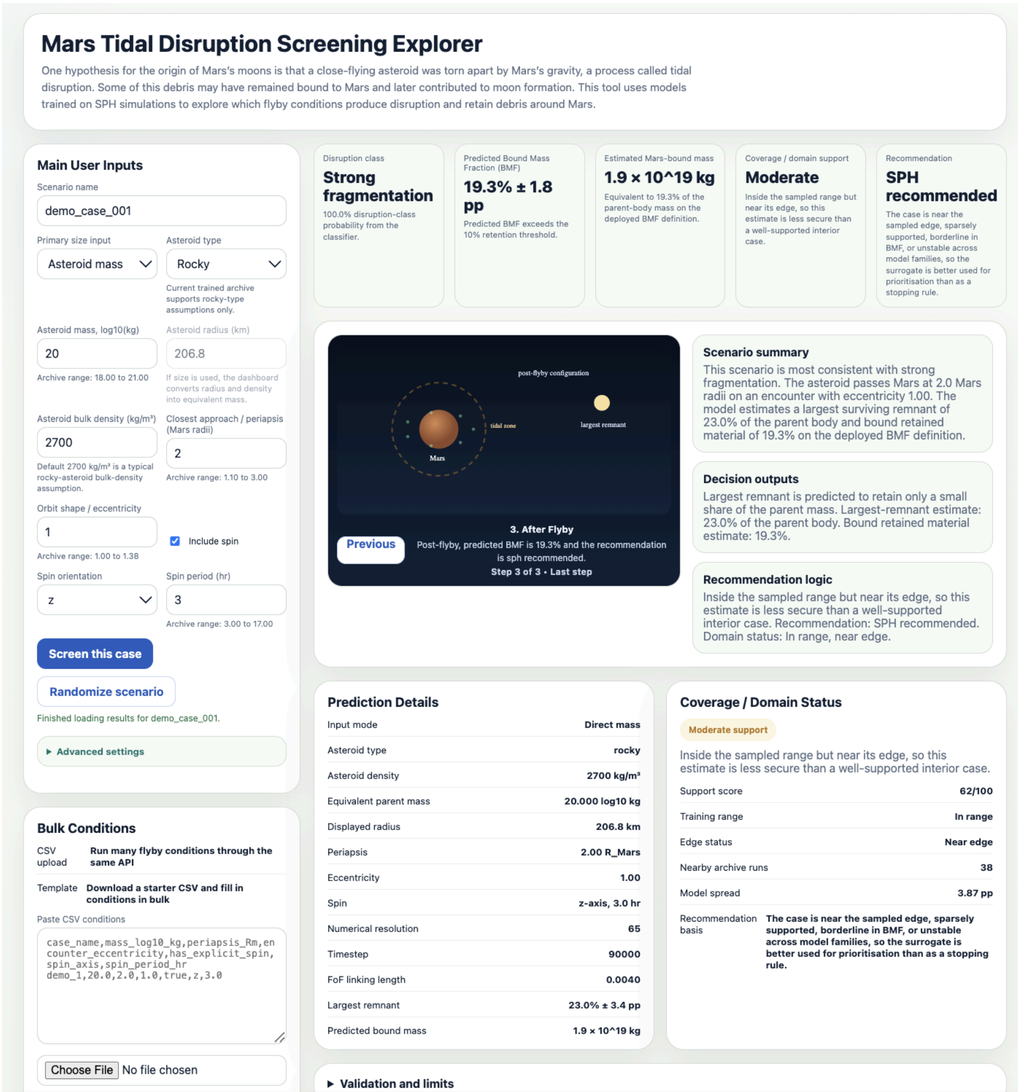

Figure A1. Mars Tidal Disruption Screening Explorer. The dashboard uses the deployed raw-input Random Forest to report predicted BMF, estimated Mars-bound mass, local-support diagnostics and an SPH follow-up recommendation; the 10% BMF threshold is a screening rule rather than evidence that a moon forms.